

Projectile

Brazil-Uddin-Uhan

$$d = vt \text{ [for constant speed]}$$

$$V_f = V_o + at$$

$$V_f^2 = V_o^2 + 2as \text{ [constant acceleration]}$$

$$d = \frac{1}{2} [V + V_f] t \Rightarrow d = v_{avg} \times t$$

$$s = V_o t + \frac{1}{2} at^2$$

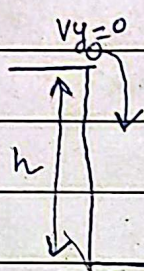
$$S = X_f - X_i$$

or vice versa

$$h = \frac{V_o^2 \sin^2 \theta}{2g}$$

$$d = V_o t + \frac{1}{2} at$$

Case 1-



So in this $V_{y0} = 0$

$$h = \frac{1}{2} at^2$$

$$R = \text{Range} \Rightarrow dx = V_x t$$

$$a_x = 0, a_y = 9.81$$

or $R = V_x t$ used to find vertical time.

$\rightarrow$ is initial velocity

To find R_{final}

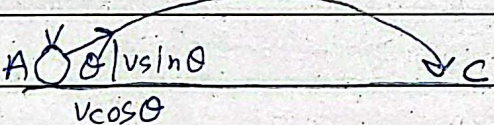
$$\text{Final Velocity} = V_{yf} = V_{y0} + gt$$

if we want to find velocity before hitting ground so $= \sqrt{V_x^2 + V_y^2}$

$$\theta = \tan^{-1} \left(\frac{V_y}{V_x} \right)$$

BC $V_y = 0$

Case 2:-



$$t_{AB} = \frac{V \sin \theta}{g}$$

$$V_f = V_o + at$$

$$0 = V \sin \theta - gt \Rightarrow t = \frac{V \sin \theta}{g}$$

$$R = V_x t$$

$$V_f^2 = V_o^2 + 2ay \Delta y$$

$$0 = (V \sin \theta)^2 - 2g h$$

$$h = \frac{V^2 \sin^2 \theta}{2g}$$

$$t_{A \rightarrow C} \rightarrow \frac{2V \sin \theta}{g}$$

$$R = \frac{V^2 \sin 2\theta}{g}$$

GENIUS

$$s = ut + \frac{1}{2} at^2$$

$$V_{avg} = \frac{u + V}{2} \rightarrow V = u + at$$

$$V = \frac{(u + V + at)}{2} \quad V = s/t$$

$$\frac{s}{t} = \frac{2u + at}{2}$$

$$V = u + at$$

sq. vol. for while

$$V^2 = u^2 + 2at + a^2 t^2$$

$$V^2 = u^2 + 2a \left(ut + \frac{1}{2} at^2 \right)$$

$$V^2 = u^2 + 2as$$

$$s = ut + \frac{1}{2} at^2$$